

SEMYACHIK, Russian Federation

WMO: 325090

Lat: 54.0822N

Lon: 159.9853E

Elev: 89

StdP: 14.65

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.0	9.8	-12.1	2.9	10.1	-8.4	3.5	12.3	53.5	26.4	45.4	25.8	11.3	320	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	68.4	58.2	64.8	57.5	61.8	56.2	60.9	65.2	59.0	62.8	57.4	60.5	10.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.1	75.1	62.8	57.4	70.5	60.4	55.8	66.6	58.6	27.1	65.3	25.8	62.9	24.8	60.7	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.0	33.8	28.6	2.8	77.1	5.5	5.6	-1.2	81.1	-4.4	84.4	-7.4	87.5	-11.4	91.6		
(5)			1.3	64.8	4.8	3.0	-2.2	66.9	-5.0	68.7	-7.7	70.4	-11.2	72.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	21.4	21.6	25.4	31.3	38.9	46.8	53.8	56.1	51.2	42.1	30.9	23.7	(6)
(7)		DBStd	13.40	6.51	6.52	5.36	4.05	4.28	5.03	4.68	4.07	4.35	4.93	5.60	5.99	(7)
(8)		HDD50	5162	886	795	763	562	346	123	14	1	35	248	573	816	(8)
(9)		HDD65	10220	1351	1215	1228	1012	809	546	350	280	414	711	1023	1281	(9)
(10)		CDD50	422	0	0	0	0	1	27	131	189	72	2	0	0	(10)
(11)		CDD65	5	0	0	0	0	0	0	2	3	0	0	0	0	(11)
(12)		CDH74	32	0	0	0	0	0	4	17	11	0	0	0	0	(12)
(13)		CDH80	6	0	0	0	0	0	0	5	1	0	0	0	0	(13)
(14)	Wind	WSAvg	10.5	14.6	15.2	13.1	9.8	6.8	6.5	5.4	6.6	9.1	11.0	13.1	14.9	(14)
(15)	Precipitation	PrecAvg	47.7	3.6	3.6	4.4	4.3	2.8	3.5	3.1	4.2	3.9	5.4	4.7	3.9	(15)
(16)		PrecMax	68.4	8.4	14.6	10.6	14.2	7.9	10.3	8.0	8.1	8.7	12.9	14.0	13.1	(16)
(17)		PrecMin	30.9	0.4	0.6	0.9	0.7	0.8	0.2	0.2	0.7	1.1	0.8	0.9	0.5	(17)
(18)		PrecStd	8.6	2.0	3.0	2.1	3.0	1.6	2.4	1.9	2.2	1.8	3.3	2.9	3.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	34.9	37.0	47.0	59.4	70.8	74.8	74.8	68.0	56.8	44.0	38.5	(19)
(20)			MCWB	32.4	31.5	31.9	37.8	46.7	55.8	61.2	59.8	56.0	46.3	39.0	36.9	(20)
(21)		2%	DB	33.7	32.6	34.4	41.4	52.2	62.0	67.5	68.7	63.4	52.9	42.0	36.4	(21)
(22)			MCWB	31.8	30.7	30.5	34.6	43.8	53.0	59.6	58.8	54.1	44.2	39.3	34.6	(22)
(23)		5%	DB	32.5	31.4	32.8	38.6	48.4	57.7	63.7	65.1	60.2	50.4	40.4	34.2	(23)
(24)			MCWB	30.9	29.8	30.0	33.4	42.2	51.1	58.8	57.9	52.4	43.9	37.9	32.0	(24)
(25)		10%	DB	31.3	30.4	31.6	36.6	45.4	53.9	60.6	62.1	57.6	48.6	38.9	32.5	(25)
(26)			MCWB	29.7	28.7	29.7	32.4	40.8	49.6	57.2	57.1	51.4	43.1	36.0	30.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	32.5	33.6	39.2	48.0	57.6	63.7	64.0	59.4	49.5	42.3	37.5	(27)
(28)			MCD B	34.2	33.1	35.2	44.8	56.1	66.5	70.5	69.7	63.6	51.7	42.7	37.9	(28)
(29)		2%	WB	32.4	31.5	32.1	36.0	45.1	54.3	61.1	61.1	56.0	47.7	40.4	35.3	(29)
(30)			MCD B	33.0	32.2	32.9	39.3	50.3	60.5	65.6	65.3	60.4	50.0	41.1	36.4	(30)
(31)		5%	WB	31.5	30.6	31.3	34.5	43.1	51.9	59.1	59.4	54.4	46.2	38.7	32.9	(31)
(32)			MCD B	32.3	31.4	32.1	37.1	46.9	56.3	62.8	63.0	58.0	48.4	39.8	33.7	(32)
(33)		10%	WB	30.0	29.1	30.4	33.3	41.4	50.0	57.3	57.9	52.9	44.6	36.8	31.1	(33)
(34)			MCD B	31.2	30.2	31.3	35.4	44.6	53.1	60.3	61.0	56.0	46.9	38.4	32.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	7.0	7.2	7.2	8.1	7.9	7.6	7.9	9.8	8.7	7.4	6.7	(35)
(36)			MCDBR	4.8	5.2	7.1	10.2	14.6	16.4	15.0	15.0	14.2	11.0	7.8	5.9	(36)
(37)			MCWBR	4.9	5.3	5.9	6.9	9.0	9.1	8.4	8.4	8.5	7.8	7.7	6.6	(37)
(38)		5% WB	MCD BR	4.5	4.9	5.2	8.7	12.7	13.6	12.5	11.8	11.6	8.6	6.7	5.9	(38)
(39)			MCWBR	5.1	5.4	5.2	6.8	8.9	8.6	7.9	8.0	8.6	8.0	8.0	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.322	0.374	0.423	0.479	0.466	0.475	0.446	0.367	0.348	0.305	0.283	(40)	
(41)		taud	2.130	2.110	1.973	1.904	1.855	2.076	2.099	2.178	2.396	2.287	2.184	2.111	(41)	
(42)		Ebn at Noon	202	236	246	251	243	250	244	244	254	227	195	177	(42)	
(43)		Edh at Noon	21	30	44	54	60	49	47	41	29	26	20	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	246	486	911	1393	1568	1629	1557	1254	1016	601	311	190	(44)	
(45)		RadStd	27	67	92	139	171	157	128	117	90	57	28	22	45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page